Analysis & Strategy for Portuguese Hotel Enterprise

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1 Introduction

Our consultation target is a major Portuguese hotel enterprise faces a critically high booking cancellation rate, leading to significant lost revenue and operational unpredictability. Our objective is to analyze the provided booking data to identify the root causes of cancellations, using data-driven insights and recommendations to formulate an effective development strategy aimed at stabilizing bookings and maximizing overall profitability.We propose to use the Power BI dashboard to track and analyze booking trends, cancellation drivers and potential losses in real time through interactive visualizations and key performance indicators (KPIs), thereby supporting management in making rapid, data-driven decisions. The expected benefits include a significant reduction in booking cancellation rates and enabling hotels to conduct revenue management and customer segmentation more accurately.

2 Data Summary

This dataset comes from Kaggle (an open data portal), specifically the Hotel Booking Demand dataset. It contains booking information for 2 hotels (City Hotel & Resort Hotel) spanning 2015–2017, including details of individual bookings (e.g., guest count, booking status, room type).

Data dictionary can be found in the Appendix B.

3 Data Processing and Transformation

During the initial phase, we have some missing values in columns like babies, children, agent and company, causing errors in data loading. Furthermore, since there are lots of people cancel their bookings, we have to distinguish the successful bookings and canceled bookings to analyze the cancellation rate.

To address these issues and prepare the data for modeling, the following transformation steps were executed in Power Query:

- Missing value handling: Set children error values to 0; uniformly label missing agent/company fields as 'Unused' and country fields as 'Unknown'.
- Feature engineering: Create a unified arrival_date based on year, month, and day, while generating numerical month formats.
- Structural optimization: Remove 15 redundant columns not used for core analysis to streamline the fact sheet.

4 Data Modeling

4.1 Model Creation

This project adopts a star schema to structure data for efficient analysis and dashboard interactivity, consisting of one fact table and two dimension tables (as visualized in the model diagram).

Table-Specific Configuration:

1. Dim_Date Generated using the DAX CALENDARAUTO() function, which automatically creates a date range aligned with the dataset (excluding 2014 to match the 2015–2017 booking data).

   Contains attributes like DayOfWeekName, Month, and Year to support granular time-based filtering (e.g., monthly cancellation trends).

2. Dim_Country Processes country codes from the fact table to map each country to its corresponding continent (e.g., grouping guest origins by region).Configured by setting the data category of the country field to "Country/Region" (for geographic visualization support, e.g., map charts of booking origins).

This star schema design ensures fast data retrieval and flexible filtering—critical for building interactive dashboards that analyze booking/cancellation trends across time, geography, and other dimensions.

4.2 DAX

1. Calculated Columns

   These columns are used to create new dimension attributes in the fact table (hotel_bookings) for flexible grouping and slicing.

DAX Name	**Type**	**Code (Logic)**	**Purpose**
Average Daily Rate Group	Calculated Column	ADR grouped into: 0–100, 100–200, 200–300, and above 300	Used to analyze pricing segments and compare performance across different ADR levels.
Lead Time Group	Calculated Column	Lead time grouped into: 0–7 days, 8–30 days, 31–90 days, and above 90 days	Used to study advance booking behavior and its impact on cancellation risk.

Table 1: Calculated Column

2. Base Aggregations

   These core metrics for counting and summing form the foundation for building all complex KPIs in the model.

DAX Name	**Type**	**Code (Logic)**	**Purpose**
Total Bookings	Measure	Counts all booking records	Used to measure overall demand volume.
Stayed Nights	Measure	Sums weekday and weekend nights for non-canceled bookings	Used to calculate actual room occupancy.
Total Revenue	Measure	ADR multiplied by total stayed nights for non-canceled bookings	Used to evaluate realized revenue performance.

DAX Name	Type	Code (Logic)	Purpose
Lost Revenue	Measure	ADR multiplied by total nights for canceled bookings	Used to estimate potential revenue loss and identify high-risk segments.

Table 2: Base Aggregations

3. Derived KPIs and Ratios

   These metrics filter and calculate basic aggregated values to provide key performance indicators (KPIs)

DAX Name	Type	Code (Logic)	Purpose
Stayed Bookings	Measure	Counts bookings with `is_canceled = 0`	Used to measure actual check-in volume.
Canceled Bookings	Measure	Counts bookings with `is_canceled = 1`	Used to quantify the total number of cancellations.
Cancellation Rate	Measure	Canceled Bookings divided by Total Bookings	Core risk indicator used to assess booking instability.
Cancellation Rate Benchmark	Measure	Fixed benchmark value set at 40%	Used as a reference line to compare actual cancellation rates.
Avg ADR (Stayed)	Measure	Total Revenue divided by Stayed Nights	Used to evaluate pricing performance for occupied bookings.

Table 3: Derived KPIs and Ratios

4. Contextual Analysis

These metrics are used to compare and rank across dimensions (e.g., countries/regions) to provide business insights.

DAX Name	Type	Code (Logic)	Purpose
Revenue Contribution	Measure	Calculates Total Revenue in current filter context divided by total revenue of all countries	Used to identify key markets and their share of total revenue.
Rank	Measure	Ranks countries by Total Revenue (highest to lowest)	Used to compare performance and highlight top-performing markets.

Table 4: Contextual Analysis

5 Dashboard Design and Visualizations

Diagrams can be found in the Appendix A.

5.1 Dashboard Structure Overview

The Power BI dashboard is structured into four analytical sections: Overview, Revenue Analysis, Cancellation Analysis, and Income and Risk Analysis.

1. The Overview section provides a concise snapshot of overall hotel performance, enabling users to quickly understand key operational conditions.

2. The Revenue Analysis section examines revenue trends and profitability drivers across hotel types and time periods.

3. The Cancellation Analysis section focuses on identifying the key factors influencing booking cancellations, particularly for city hotels.

4. Finally, the Income and Risk Analysis section integrates average daily rate (ADR) and cancellation rate to support balanced risk assessment and decision-making.

5.2 Overview

This overview page provides a snapshot of hotel operations. It focuses on core metrics such as transaction volume, financial performance, and overall risk levels.

5.2.1 Key Visualizations & Rationale

1. Cards & Waterfall Chart

    - Visual:

      Overview of the corporation

    - Rationale:

      To begin with, we use some cards to show the key metrics, since it can highlight the key business indicators in the most concise way.

      Meanwhile, to show the influence of each hotel on the overall outcome, we also use the waterfall chart to show the total revenue by hotel type.

2. Area Chart

    - Visual:

      Monthly trends of bookings and cancellations

    - Rationale:

      We apply Area Chart to illustrate the monthly trends of bookings and cancellations. This combined chart was chosen for the dual-axis time series analysis, which enables users to simultaneously observe the monthly trends and synchronicity of two related but significantly different indicators on the same time axis.

3. Line and Clustered Column Chart

    - Visual:

      Time series relationship of financial indicators

    - Rationale:

To present the time series relationship of financial indicators, we use a line and clustered column chart, which shows the time series trends of income and average housing prices.

4. Clustered Bar Chart

   - Visual:

     Comparison of the contributions of different customer types

   - Rationale:

     We use a clustered bar chart to compare the contributions of different customer types to actual revenue and potential lost revenue, because it is the best choice for comparing the magnitudes of values across different categories

5. Stacked Bar Chart & ArcGIS

   - Visual:

     Revenue contribution from different countries

   - Rationale:

     Finally, we adopt a stacked bar chart to analyze the revenue contribution from different countries since it is a good method for comparison of revenue contribution between different countries. To be more specifically, we use ArcGIS in PowerBI and a table in dashboard 2 to demonstrate more detail information of customers from different countries.

5.3 Revenue Analysis

5.3.1 Overview

In this section, dashboards are designed to demonstrate an overall management overview by virtue of revenue analysis. The first dashboard of this section includes the pulse of corporation revenue which focuses on the City Hotels' performance and detailed breakdown of revenue streams. The second section contains the profitability drivers by right of the key influence analysis.

5.3.2 Key Visualization & Rationale

1. Line Chart & Line and Clustered Column Chart

   - Visual:

     (a) Monthly Revenue Trend

     (b) Stayed Booking and Avg ADR by Hotel

   - Rationale:

     Initially, with overview revenue overview of two different hotels, we can find out which hotel is our ”long board” (advantage) roughly. Then, the specific price level and performance should be analyzed in order to identify the ”long board” accurately.

2. Decomposition Tree

   - Visual:

     Revenue Stream Breakdown

   - Rationale:

     This visual is designed to gain the root cause analysis to obtain the the cash flow direction. In this way, how to improving the total of revenue can be found. Therefore, the revenue volume drivers can be obtained.

3. Key Influencers

   - Visual:

     Key Drivers for Avg ADR

   - Rationale:

     `country`, `distribution_channel` and `customer_type` have been selected as explanatory factors because they are identified as high-volume revenue drivers in the decomposition tree. Hence, we intend to figure out if these volume factors are also the efficiency revenue drivers. Meanwhile, `total_of_special_requests` is introduced to analyze the behavior dimension.

5.4 Cancellation Analysis

5.4.1 Overview

Based on our previous analysis of hotel revenue, we have identified city hotels as our main type of promotion. In the fifth and sixth dashboards of this study, we will analyze the factors that affect the cancellation rate of city hotel bookings. By visualizing and analyzing the driving factors, we can identify the categories, demographics, or time factors that have the greatest impact on hotel cancellation rates, and gain insights into the common characteristics of high cancellation rates and high certainty customer sources.

5.4.2 Key Visualization & Rationale

1. Key Influencers

   - Visual:

     Key Drivers for is_canceled

   - Rationale:

     To screen the top few factors that have the strongest impact on is_canceled from multiple dimensions such as ADR and booking channels, helping to quickly identify high-risk dimensions that need priority intervention and providing data support for subsequent in-depth analysis of the top influencing factors.

2. Dashboard 6

   On page six, we focused on designing six visualization charts, providing targeted visualization designs for five important analysis factors, making high-risk classification clear at a glance and providing intuitive support for subsequent in-depth analysis and strategic planning.

   (a) Vertical Bar Chart

   - Visual:

     Average Daily Rate (ADR) vs Cancellation Rate Chart

- Rationale:

  Select the ascending vertical bar chart, use height comparison to directly reflect the cancellation rate, and understand the high-risk cancellation rate range by classifying ADR by interval.

(b) Horizontal bar chart

- Visual:

  Distribution channel and cancellation rate

- Rationale:

  Using a horizontal bar chart, we can intuitively compare the cancellation rates of four distribution channels, analyze channels with cancellation rates exceeding the normal range, assess the quality of channels and customers based on their cancellation rates, and provide targeted feedback and optimizations according to the cancellation habits of different channels.

(c) Funnel chart

- Visual:

  Customer type & Cancellation rate

- Rationale:

  Visually present the differences in cancellation rates among various types of customers, quickly understand the core risk customer groups as well as high quality customer types, introduce different improvement strategies for different customer groups.

(d) Line chart

- Visual:

  Advance booking Period & Cancellation rate

- Rationale:

  By presenting the trend changes between advance booking period and cancellation rate through a line chart, it is possible to intuitively see the trend of cancellation rate changes. By understanding the high-risk attributes of ”long-term

bookings", differentiated refund and change policies can be formulated.

(e) Circular chart

- Visual:

  Repeated & Cancellation Rate

- Rationale:

  Compared to pie charts, circular charts provide a more concise analysis of two categories, using industry labels to indicate cancellation rates and comparing the proportion of new and returning customers within the circle.

5.5 Income And Risk Analysis

5.5.1 Overview

After conducting a detailed analysis and providing suggestions for the cancellation rate and revenue situation of the city hotel, we need to conduct a comprehensive analysis that combines the risks represented by the cancellation rate with the revenue ability represented by average ADR. Currently, the benchmark for cancellation rate in the hotel industry is approximately 40% (Hertzfeld, 2019). This part serves as a supplement to the utilization of the "strengths-based" effect. In this case, we can avoid the risk of a monotonous structure and propose diversified risk mitigation strategies.

5.5.2 Key Visualization & Rationale

1. Gauge

   Gauge is used to compare the cancellation rates of different categories with the hotel industry benchmark cancellation rates.

2. Bubble diagrams & Bar charts

   We use bubble diagrams to analyze the relationship between cancellation rate and average ADR (Stayed), according to customer type, distribution channel, reserved room type and continent. And the slicer enables the operator to view the specific relationship of different quarters.

3. Q&A Box

   Using the Q&A box, users can pose natural language questions and receive answers in the form of visual objects.

4. Bar charts

   The bar chart is used to observe the detail cancellation rate of different continents.

6 Analysis and Insights

6.1 Overview

6.1.1 Overview of the corporation

1. The total bookings of the whole hotel enterprise is 119,000 in 2015-2017 and the total revenue is €26 million.
2. The potential lost revenue is €16.73 million and the average daily rate is €101.9 per room.
3. The cancellation rate is 37.04%
4. The total revenue of City Hotel is approximately €14million, the total revenue of Resort Hotel is €12 million, and the total revenue of the whole enterprise is €26 million.

6.1.2 Monthly trends of bookings and cancellations

This combined chart was chosen for the dual-axis time series analysis, which enables users to simultaneously observe the monthly trends and synchronicity of two related but significantly different indicators on the same time axis. From the chart we can find out the total booking volume fluctuates between 2K and 6K and the cancellation rate not fully synchronized with the booking volume fluctuations.

6.1.3 Time series relationship of financial indicators

Graph shows the time series trends of income and average housing prices. It's evident that summer is the peak season for hotel bookings. And the Total Revenue is positive related to

Average ADR.

6.1.4 Comparison of the contributions of different customer types

From the chart we can see that transient customers contribute the most revenue, but also lose the most revenue.

6.1.5 Revenue contribution from different countries

It is discovered that Portugal (PRT) leads with 0.21 (21%) of the total revenue, followed by United Kingdom (GBR) at 0.16 (16%) and France (FRA) ranks third at 0.12 (12%). While other countries contribute less than 10%, emphasizing a concentration of revenue from a few top-tier nations.

From the map we could also find that revenue is heavily dependent on European countries, with Portugal being the dominant contributor (accounting for over 20%). The combined revenue contribution from the top three countries approaches 50%, making them the core market for the hotel business.

In addition, we add a drill-through of country in Top 10 Revenue Contribution by country circular chart (Page1) and Page 2. The drill-through feature allows the user to automatically transfer the data of a selected country/region from the overview page (Stacked Bar Chart) to the details page (Page 2) for in-depth analysis via tables and maps.

6.2 Revenue Analysis

6.2.1 Strategic Positioning: The ”Long Board” Effect

Our comparative analysis indicates that a distinct strategic consensus: Although resort hotel performs in a powerful position during the festival period (July to August). The city hotel is still the main strength of the portfolio.

”Monthly Revenue Trend” reveals that resort hotel could bring more revenue during the festival period. From 2015 to 2017, city hotel handled 46k orders and the ADR is €107 however resort hotel only tackled 29k orders and the ADR is €97. In this way, city hotel outperforms

the resort hotel in both volume and average price which illustrates a stable and high-yield baseline. Therefore, our strategy focuses on expanding dominance rather than fixing the resort's seasonality.

6.2.2 Revenue Stream Breakdown

This section utilizes the decomposition tree to indicate that revenue of city hotel is driven by intermediaries. Over 85% of revenue flow is from TA/TO which is tied to the travel agency. Therefore, we are not to give up TA/TO pattern but optimize the kind of customer in this model. Data demonstrates that customer position of city hotel is mainly centered in the transient field from Western Europe which company should pay attention to this information.

6.2.3 Profitability Drivers

Despite analyzing previous revenue breakdown, we intend to improve the ADR based on a certain revenue baseline. Thus, key influencers visual furnishes a sharp conclusion: only two factors which is `customer_type` and `total_of_special_requests` statiscally significant in driving up ADR.

1. Transient Premium (+15.55):

   When `customer_type` is Transient, avg ADR increases €15.55 which indicates that individual visitors are the most valuable assets of the city hotel. Unlike the group customers often negotiate the discounts, transient customers usually pay the market rates.

2. The Service Signal (+12.23):

   When `total_of_special_requests` is more than 1, avg ADR increases €12.23 which indicates that the customers who engage with the hotel such as asking for views are signaling a low price sensitivity.

3. The Top Segment Profile (ADR €125.4):

   The profile with Transient + Special Requests achieve a €125.4 ADR which relatively higher than overall ADR. In this way, we can conclude that valuable customer are coming from TA/TO channel, provided they are Transient and active in making requests.

Hence, we do not need to shift the channels to reinforce ADR. Instead, we should shift the behavior of guests within our existing channels.

6.3 Cancellation Analysis

6.3.1 Average daily rate vs Cancellation Rate

According to data visualization that as average daily rate rise, the cancellation rate gradually decreases. Rooms priced above 300 have the lowest cancellation rate at 0.3, while rooms priced below 100 have the highest cancellation rate at 0.43. The cancellation rate is higher than 0.4 in the range where the housing price is less than 200 yuan. From this analysis, we can infer that the sunk cost effect makes them more cautious and willing to fulfill their obligations when considering cancellation. Tthe higher the price paid, the greater the perceived 'loss' when they cancel. When Average daily rate is low, the sunk cost of booking is low, resulting in a higher cancellation rate, and vice versa.

6.3.2 Customer type vs Cancellation Rate

In city hotels, the cancellation rate of the Group is only 10%, which is the most stable and high-quality source of customers, indicating that the hotel's group booking business is well managed. Group bookings usually involve deposits, with strong planning and high commitment from organizers, so the risk of cancellation is extremely low.

The cancellation rate of contract customers is as high as 48%, reflecting the possible structural defects in the enterprise agreement framework. Traditional company agreements often emphasize room rate discounts but ignore performance constraints. Loose cancellation clauses make customers who use the agreed price to book the hotel an alternative option, increasing the probability of cancellation.

The cancellation rate of 46% for transient customers is also relatively high. In the digital booking environment, individual travelers can compare prices from multiple sources and decide whether to book or not. They can book rooms that can be cancelled for free on multiple platforms at the same time, and ultimately only choose the best option, thus increasing the cancellation

rate of reservations. This consumption behavior has evolved from a single purchase decision to a dynamic optimization process.

6.3.3 Distribution channel vs Cancellation Rate

The data displayed on the pie chart for city hotels shows that the cancellation rates for Direct and GDS are extremely low (<20%), making them the hotel's highest quality source of customers. This indicates that guests who book directly have the highest loyalty and planning, and the official website marketing and booking experience are successful. The GDS channel plays a role in stabilizing business customer sources. The cancellation rate of Corporate (23.1%) is at an acceptable moderate level, which meets the characteristics of business travel with certain variables. The cancellation rate of TA/TO (travel agencies) is as high as 45%, which means that nearly half of the bookings made through travel agencies will be cancelled. Exceeding the normal cancellation rate range reflects significant instability in the customer sources obtained through travel agency channels. In addition, customers who book through travel agencies tend to compare prices, services, discounts, and other aspects synchronously across different travel agency platforms, which increases the overall cancellation probability.

6.3.4 Advance Booking period vs cancellation rate

According to the data displayed by city hotels, the longer the advance booking time, the higher the probability of cancellation. The lowest cancellation rate for guests who booked within one week before check-in is 0.12. This indicates that their travel plans are very definite, usually for business trips, urgent matters with very little variability. The cancellation rate of reservations within 8-30 days is slowly increasing, and although these customers have plans, they still have enough time to cancel due to personal matters, finding better options, or changing their minds. 31-90 days & >90 days are the part with the highest cancellation rate. Guests who book months in advance face the greatest uncertainty in their schedule and may cancel their bookings due to changes in plans, prices, and other reasons.

6.3.5 Repeated vs Cancellation Rate

The cancellation rate for returning guests in urban hotels is 0.22, while the cancellation rate for non-repeated guests is 0.42. The cancellation rate of repeat customers is much lower than that of new customers. We can infer that customers will establish trust in the quality of hotel services after their first stay and form a satisfactory experience, thereby reducing price comparison and decision-making hesitation in subsequent bookings, and directly reducing their willingness to cancel. And first-time guests will be compared in multiple ways, making their booking behavior more susceptible to external alternative options, which increases the overall cancellation rate. Therefore, the first-time guest experience is crucial, and ensuring that every new guest has a unexpected experience is the key to converting them into low cancellation rate repeat customers.

6.4 Income And Risk Analysis

6.4.1 Classify according to customer type

Customers of the Transient type contribute the most to the hotel's revenue, but at the same time, their cancellation rate is also the relatively highest (higher than the 40% benchmark); The risks of both the Transient-Party and Contract types of customers are moderate, and their contribution to the hotel's revenue is relatively considerable; Although customers of the Group type have lower risks, their contribution to income is relatively smaller.

6.4.2 Classify according to distribution channel

TA/TO has the highest risk (higher than the 40% benchmark), but it has moderate income potential and it contributes the most to the hotel's revenue. While Direct and GDS have lower risks and strong revenue-generating capabilities, their contribution to the hotel's revenue is relatively weak. As for Corporate, this channel has lower risks and lower revenue potential, but it still contributes positively to the hotel's revenue.

6.4.3 Classify according to reserved room type

- High price & High risk (higher than the 40% benchmark): Room H

- High price & Low risk: Room C, F, G, L
- Low price & Low risk: Room B, D, E
- Low price & High risk: Room A

The cancellation rate of Class A room types is the highest, and their revenue-generating capacity is relatively low. However, due to the large number of reservations, they also contribute the most to the hotel's revenue.

6.4.4 Analysis of income and risks for continent - Immigration Problem

While analyzing the previous content, we found that there is always a certain group of people whose cancellation rate remains consistently high. The cancellation rate of this group is not affected by seasons, deposit policies, or distribution channels. Therefore, we have taken into account the reasons for the serious immigration problem in Europe in recent years (Eurostat, 2024).

Many people increase the probability of obtaining a visa by showing proof of hotel reservations and travel orders (International Organization for Migration, 2018). Some people apply for the visa for tourism purposes, while others do so to stay in the local area.

As a member of the Schengen Area, Portugal is inevitably confronted with the problem that some people use hotel reservations to apply for visas rather than actually intending to stay in the hotels. This will harm the interests of hotels and are detrimental to their operations and strategy making.

By using bubble charts and bar charts, I observed the situation of orders without deposit: The cancellation rate of African was relatively the highest (higher than the 40% benchmark), and it contributed less to the hotel's revenue, and the revenue potential was not high either. The slicer enables the operator to view the specific relationship of different quarters.

This also tests our hypothesis: In the absence of any cost of canceling the order, many Africans might increase their chances of obtaining a Schengen visa by providing proof of hotel reservations. Studies also show that the issue of African immigrants in European countries has become increasingly serious over the years (International Organization for Migration, 2024).

7 Recommendations and Implications

7.1 Revenue Analysis

1. Transient First Inventory Management

   Since Transient customers provide + €15.55 per night compared to others, and they are the main revenue improving driver. Company should prioritize availability for Transient guests to maximize the Transient Premium.

2. Improve Special Service Quality

   Special Requests is a verified signal of high spending power (+€12.23). Therefore, corporation should stop treating special requests as operational burdens. Instead, flag these bookings in the management system. Train front-desk staff to target these specific guests for paid upgrades upon check-in.

3. Optimize Booking Channel

   The top segment profile provides that valuable consumers usually come from TA/TO. In this way, company should focus on the dynamic pricing on the platform to ensure this segment customers are kept a high level price as they have the low price elasticity.

7.2 Cancellation Analysis

7.2.1 Customer Dimension

Core objective: To screen high intention customer groups and reduce the risk of ”impulse booking” or ”temporary changes” through differentiated services and guidance for different customer types and loyalty levels.

1. Cultivate customer loyalty and convert repeat customers:

   From the analysis of repeat customers and cancellation rates, we can propose corresponding measures to increase the number of repeat customers, such as providing exclusive discounts, early check-in, room type upgrades, and other privileges for repeat customers.

For first-time guests, ensure an excellent experience by receiving a coupon or design more comprehensive loyalty programs.

2. Differentiated management of customer types

   Focus on managing Transient and Contract customers, guiding it towards more deterministic behavior through pricing and cancellation policies. It is recommended to expand the high-quality Group customer base to ensure a relatively stable income.

7.2.2 Booking and Product Dimensions

Core objective: To reduce ”foreseeable cancellation scenarios” and increase customer ”cancellation costs” through optimization of pricing strategies, booking channel, return and change policies, and product design.

1. Optimize the booking window and implement a dynamic cancellation policy:

   For hotel reservations made in advance, bookings made more than 30 days in advance can be promoted with an ”early bird price”, but with strict cancellation policies (such as ”non-refundable” or high deposit) to reduce cancellation situations; Flexible cancellation policies can be provided for orders that are close to booking within seven days to attract customers with high certainty. Alternatively, an appropriate amount of overbooking can be made up for the vacancies caused by cancellations based on historical data of cancellation rates for future bookings.

2. Optimize booking channels and promote direct bookings:

   Relevant regulations should be proposed for TA/TO channels with high cancellation rates, such as refusing to cooperate with travel agencies with high cancellation rates, or shifting TA/TO channels to direct channels. Implementing the ”Best price guarantee on the official website” which will eliminate customers' motivation to compare prices, or creating exclusive benefits on the official website such as free breakfast, etc., to enhance the attractiveness of direct bookings.

3. Reform pricing strategy:

It is necessary to ensure the value of high-priced room types and impose stricter terms on low-priced room types to screen high certainty guests. For low-priced room types with high cancellation rates, we may be able to avoid the losses caused by cancellations by setting non-cancellable reservation plans.

7.3 Income And Risk Analysis

7.3.1 Classify according to customer type

This bubble chart may indicate that our hotel may lack the capacity to accommodate large tour groups. Therefore, the hotel should consider the rational allocation of rooms. During the annual holidays, we should be able to have sufficient capacity to receive large tour groups. At the same time, we can stabilize the contribution of other customers to our hotel's revenue and reduce the probability of their cancellations by optimizing service measures such as catering, children's facilities, and pick-up services.

7.3.2 Classify according to distribution channel

The revenue potential of the Direct and GDS channels is significant. The company should optimize its own hotel website and sales services, and deepen cooperation with third-party reservation platforms of the Global Distribution System, in order to increase revenue; The corporate (enterprise channel) customers are relatively stable but have weak profitability. They can serve as a guaranteed income source, but they are not suitable for excessive reliance.

7.3.3 Analysis of income and risks for reserved room type and Q&A Box

For Room H: This type of rooms has great revenue potential, but it has high risks. Therefore, the hotel should promote this type of rooms efforts, but should also optimize the reservation policies, such as the deposit policy and the minimum stay days.

For Room C, F, G, L: These types of rooms have great revenue potential and lower risks. The hotel should pay more attention to this type of rooms, such as increasing the promotion and discount efforts, and expanding the number of this type of rooms.

For Room B, D, E: These types of rooms have relatively lower risk but have weak profitability. Hence, they can serve as a guaranteed income source in the low season, but they are not suitable to be the profit growth point.

For Room A: The cancellation rate is the highest, and their revenue-generating capacity is relatively low. However, due to the large number of reservations, they also contribute the most to the hotel's revenue. Therefore, the hotel should improve their reservation policy, such as setting the minimum number of nights for stay, setting the deadline or specific audience (returned customers) for free cancellation;

7.3.4 Analysis of income and risks for continent - Immigration Problem

Therefore, in order to enhance the operational efficiency and reduce unnecessary preparations and services, the hotel should establish a deposit system for certain groups of people. At the same time, it should collaborate with relevant government departments, share information, set up a credit list, and pay attention to those with low credit ratings.

8 Reflection & Roles

During the process of our group project, we encountered several challenges, such as data selection and processing, choosing the dashboard theme, dashboard layout, and determining the analytical perspective.

We conducted a multi-level analysis of the hotel's operational status through various visualizations and rich interactive elements (such as slicers, drill-through, and Q&A boxes). We also provided analysis and suggestions for the hotel from a unique perspective on immigration issues.

If we had more time and resources, we might conduct a more detailed analysis, such as examining the specific conditions of each quarter or season, and provide the hotel with more detailed and personalized analysis and suggestions.

9 Member contribution

1. Ding Zixiao:

   He is responsible for the data processing, modeling, and creating measures. And some of the technical challenges are solved by him, such as some complex DAX codes and the world map. And he states the overview of the hotel's operation situation.

2. Chen Zeming:

   He is responsible for the revenue analysis of the hotel. He creates some critical measures, and he proposes a novel and constructive suggestion – maximizing "Longboard Effect". Meanwhile, he determines a beautiful dashboard theme.

3. Wu Shujin:

   She is responsible for the cancellation rate of the hotel. She also creates some critical measures, and make a good front statement about the risk analysis. And she is the person to conduct the final layout of the dashboard.

4. Zhang Shuoyan:

   He is responsible for the comprehensive analysis about the relationship of revenue and risk. He analyzes the relationship from various aspects, such as customer type, distribution channel, and room type. He conducts survey about the benchmark of the cancellation rate and the immigration problems. After that, he proposes a constructive idea about the high cancellation risk with the immigration problems and prove it through visualization.

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A Appendix A: Supplementary Figures

A.1 Data Modeling

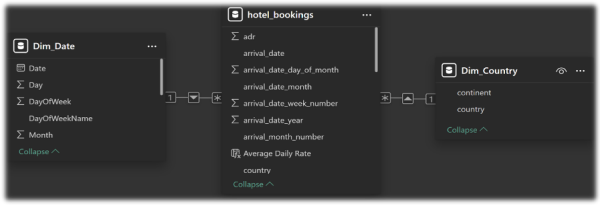


Figure A.1: Data Modeling

A.2 Dashboard Design and Visualizations

A.2.1 Dashboard Overview

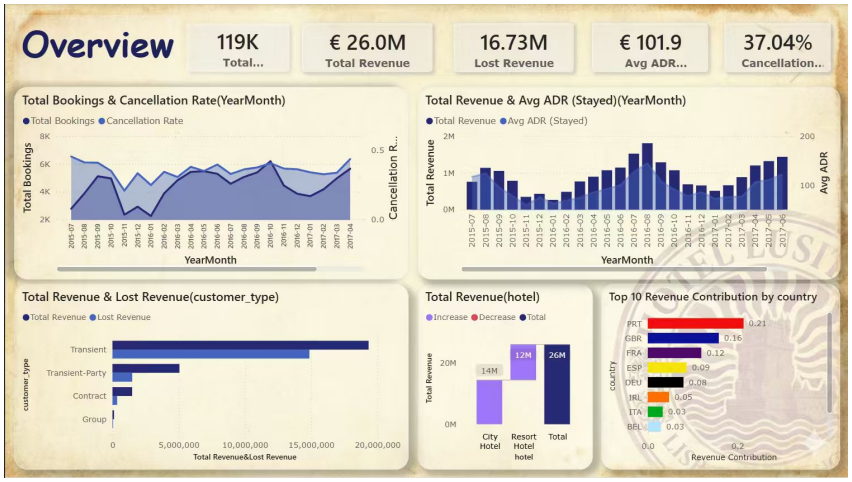


Figure A.2: Dashboard 1

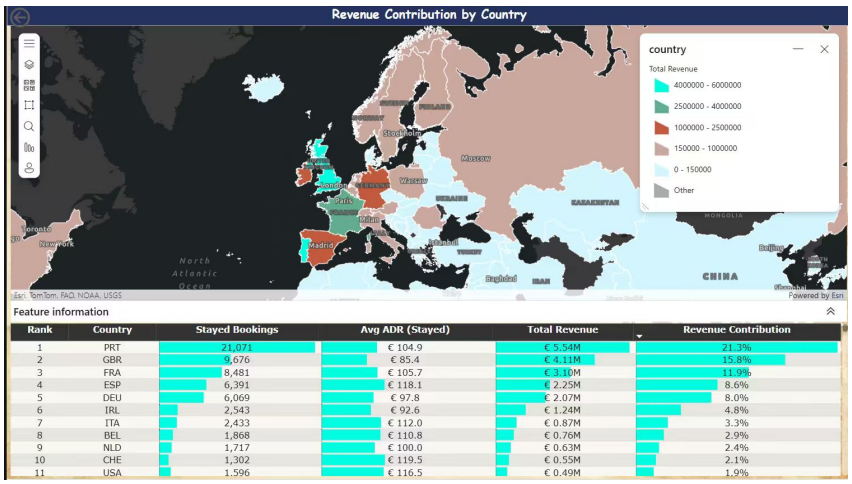


Figure A.3: Dashboard 2

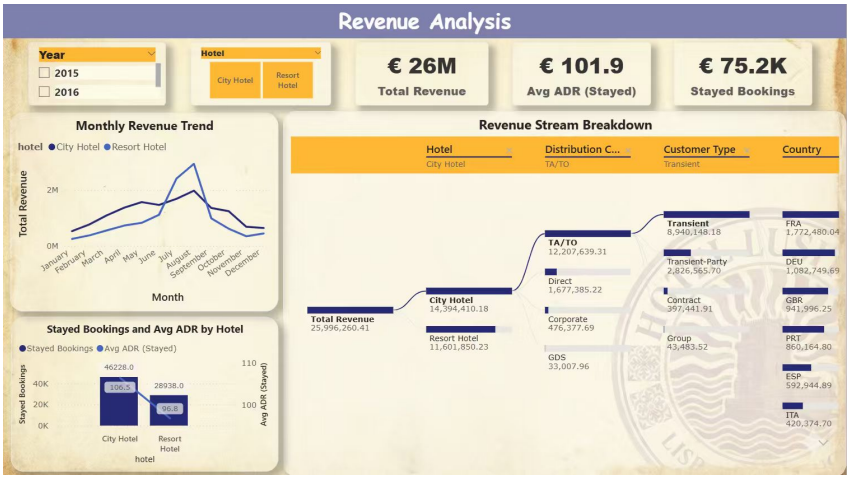


Figure A.4: Dashboard 3

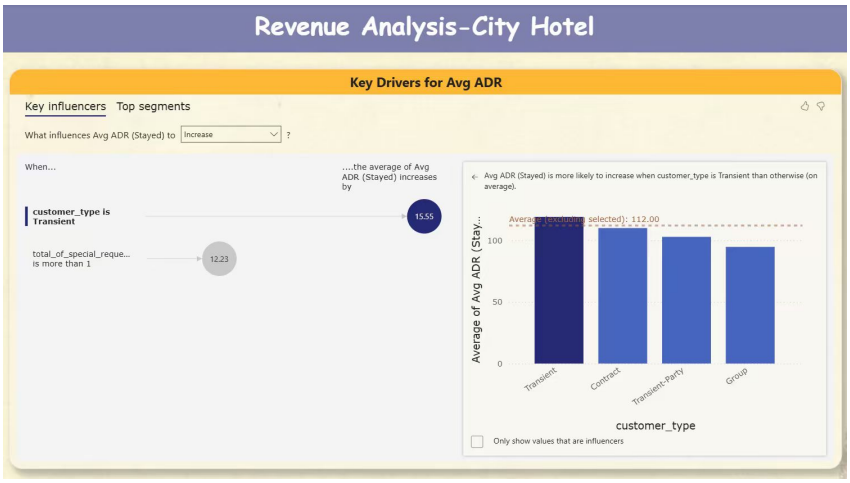


Figure A.5: Dashboard 4

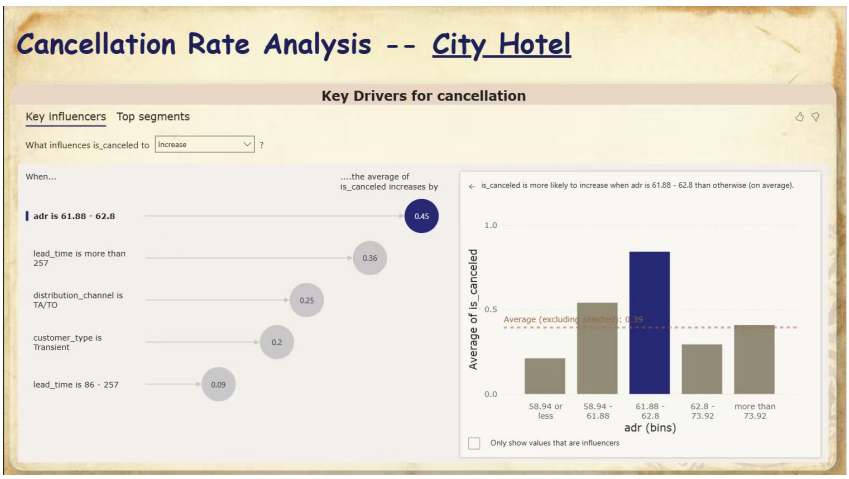


Figure A.6: Dashboard 5

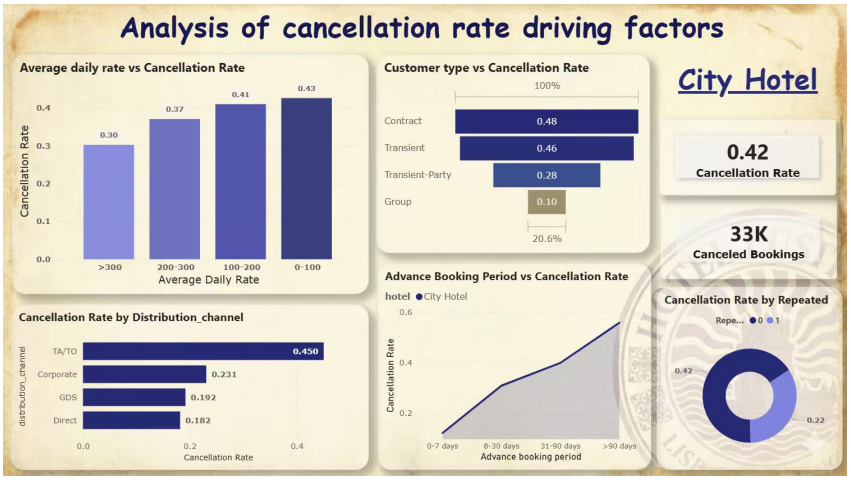


Figure A.7: Dashboard 6

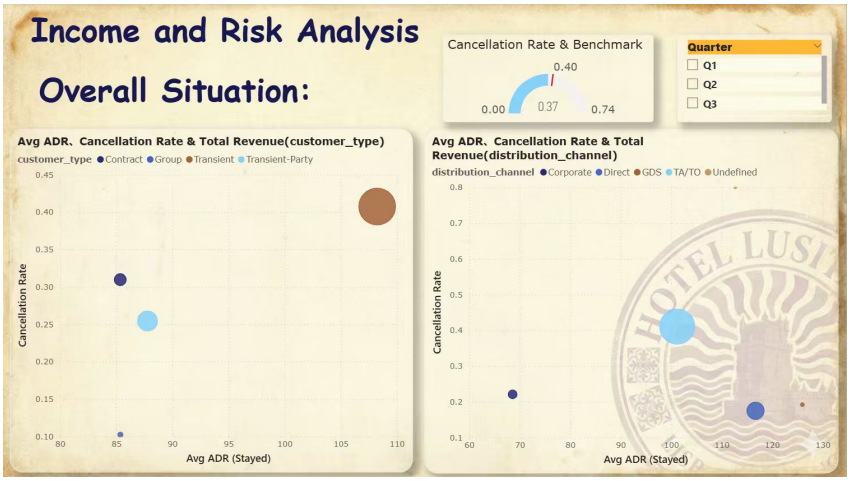


Figure A.8: Dashboard 7

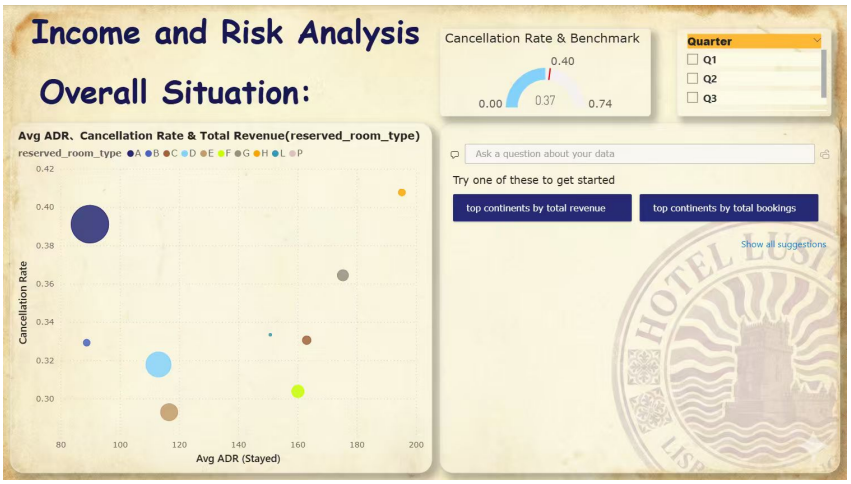


Figure A.9: Dashboard 8

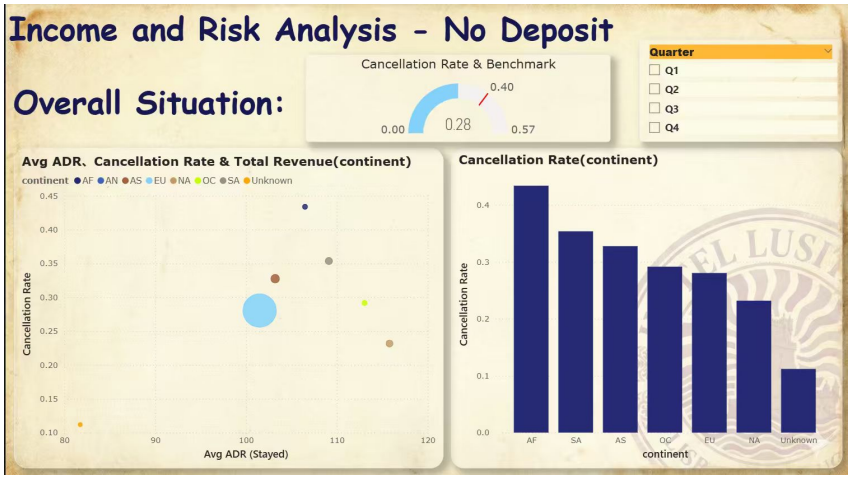


Figure A.10: Dashboard 9

A.2.2 Overview

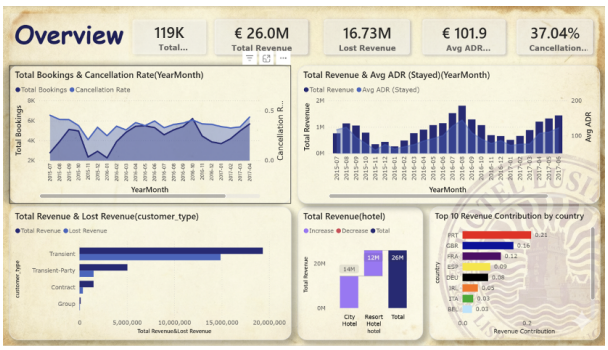


Figure A.11: Overview Dashboard

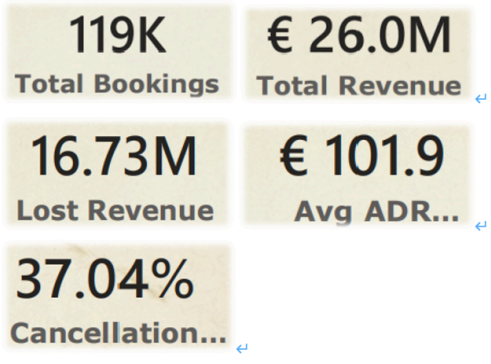


Figure A.12: Cards

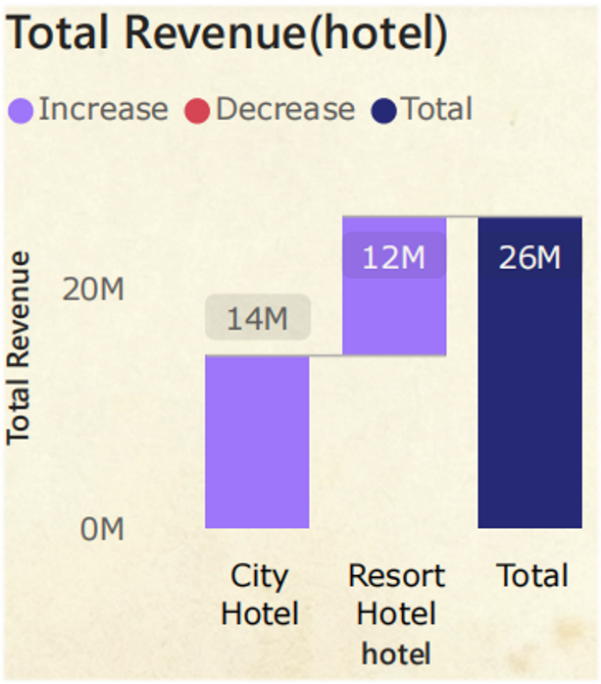


Figure A.13: Total Revenue

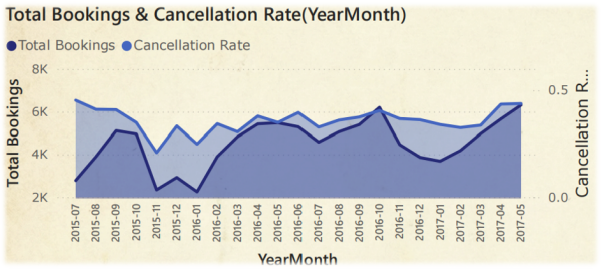


Figure A.14: Monthly trends of bookings and cancellations

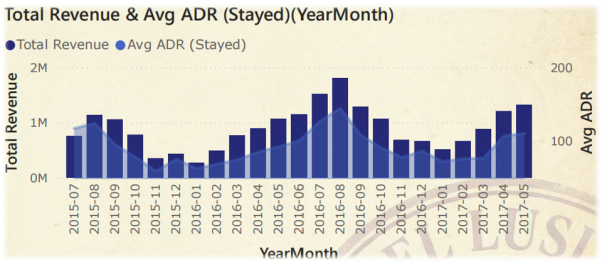


Figure A.15: Time series relationship of financial indicators

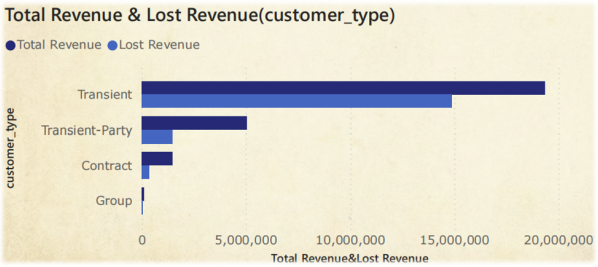


Figure A.16: Comparison the contributions of different customer types

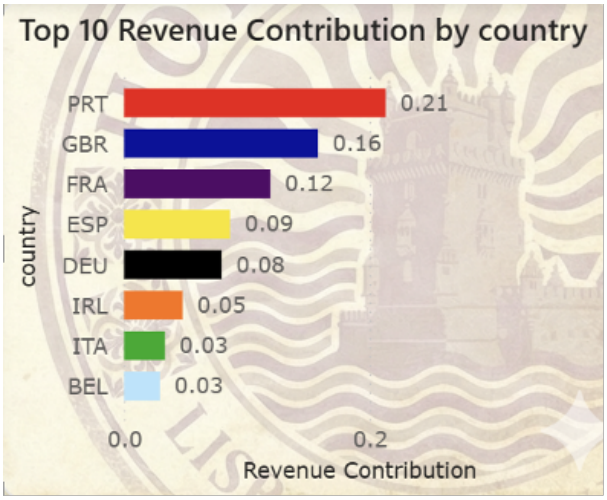


Figure A.17: Top 10 Revenue Contribution by country

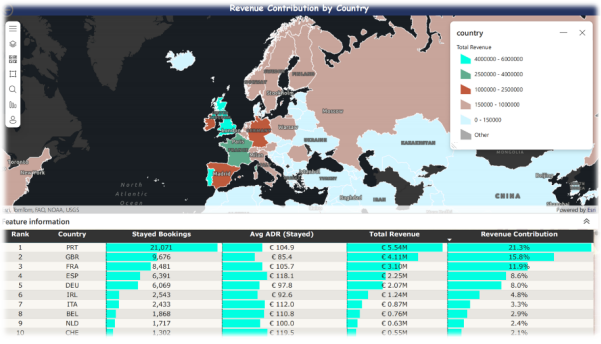
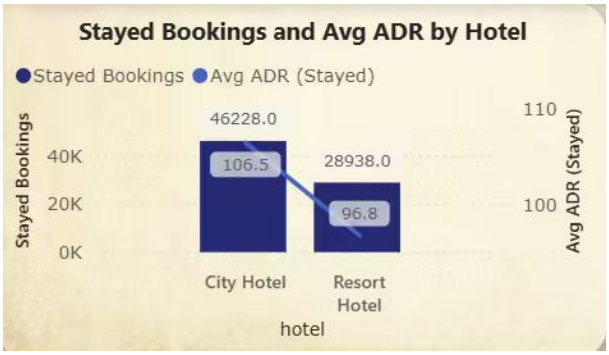


Figure A.18: ArcGIS

A.2.3 Revenue Analysis



(a) Monthly Revenue Trend



(b) Stayed Booking and Avg ADR by Hotel

Figure A.19: Line Chart & Line and Clustered Column Chart

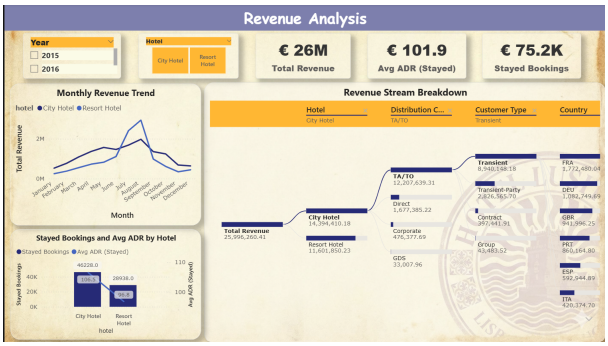


Figure A.20: Revenue Stream Breakdown

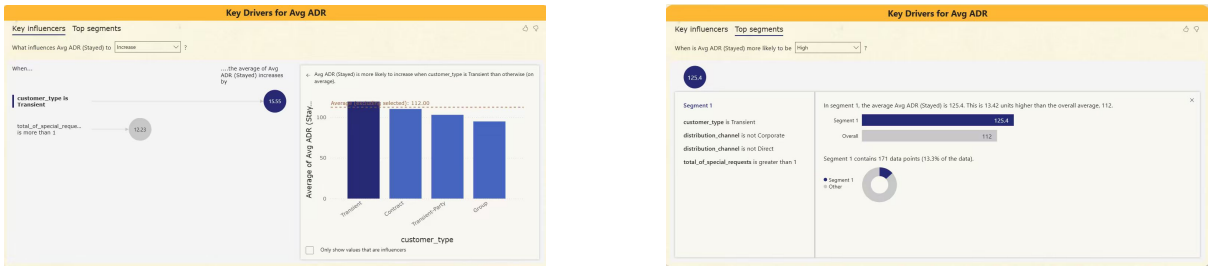


Figure A.21: Key Drivers for Avg ADR

A.2.4 Cancellation Analysis

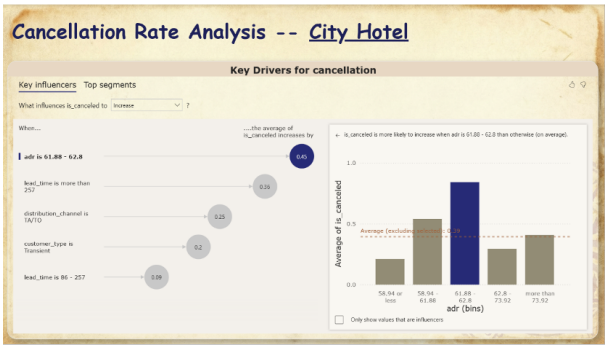


Figure A.22: Cancellation Rate Analysis

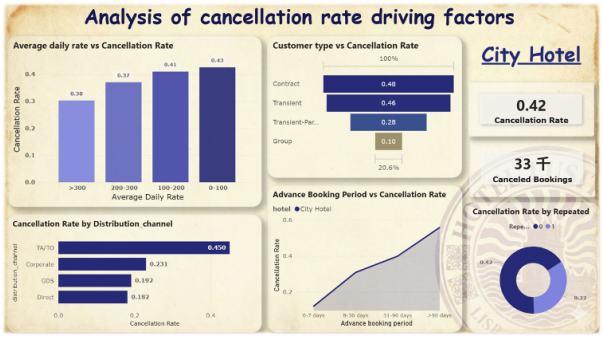


Figure A.23: Cancellation Overview

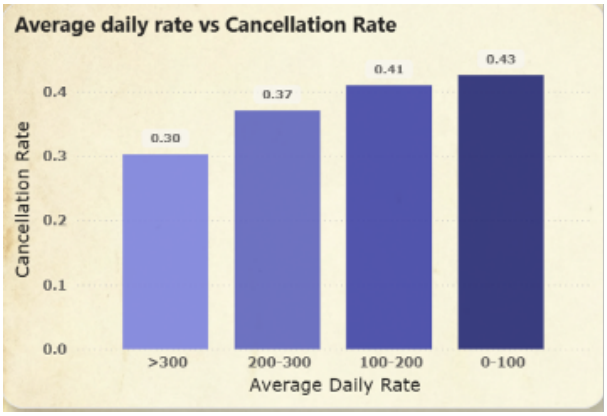


Figure A.24: Average Daily Rate (ADR) vs Cancellation Rate Chart

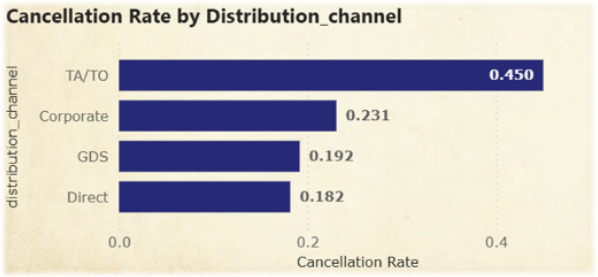


Figure A.25: Distribution channel and cancellation rate

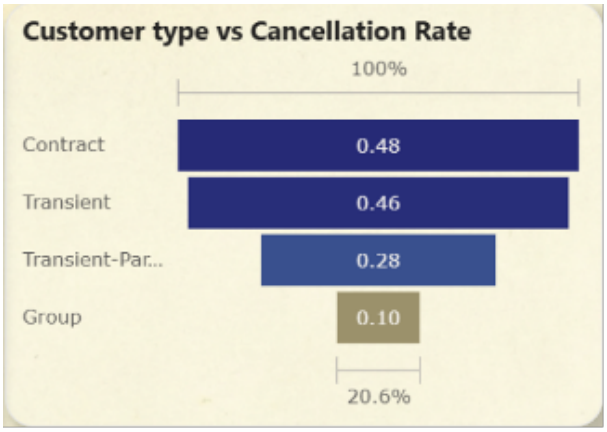


Figure A.26: Customer type & Cancellation rate



Figure A.27: Advance booking Period & Cancellation rate

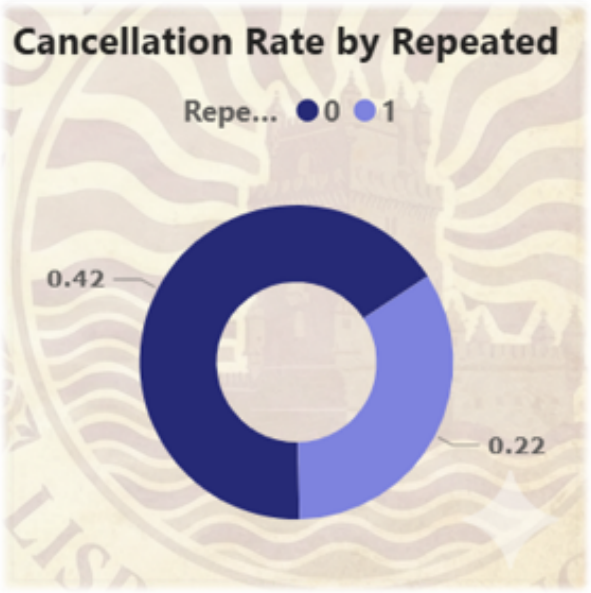


Figure A.28: Repeated & Cancellation Rate

A.2.5 Income and Risk Analysis

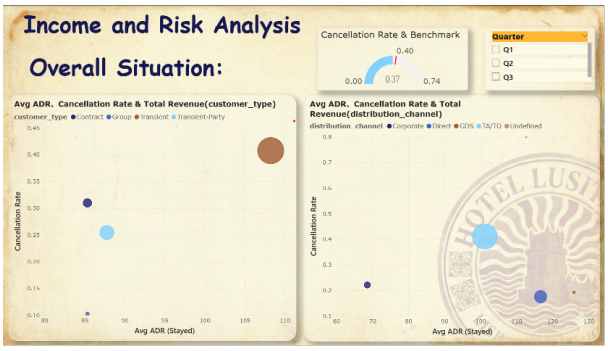


Figure A.29: Analysis of income and risks for customer type and distribution channel

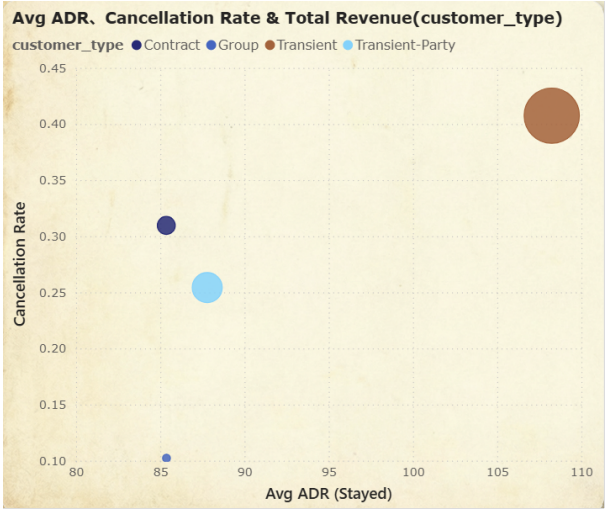


Figure A.30: Classify according to customer type

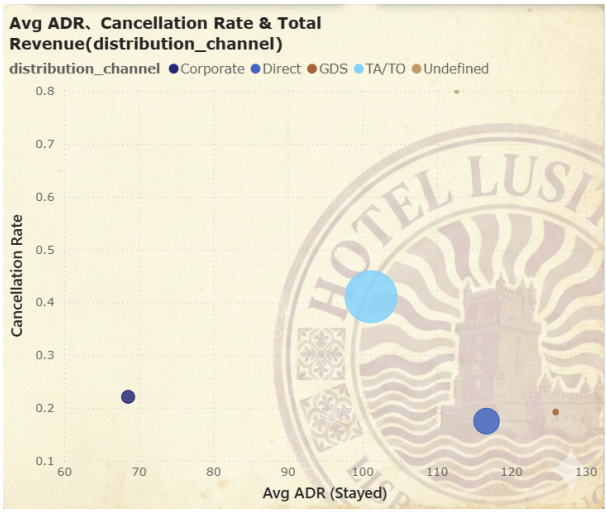


Figure A.31: Classify according to distribution channel

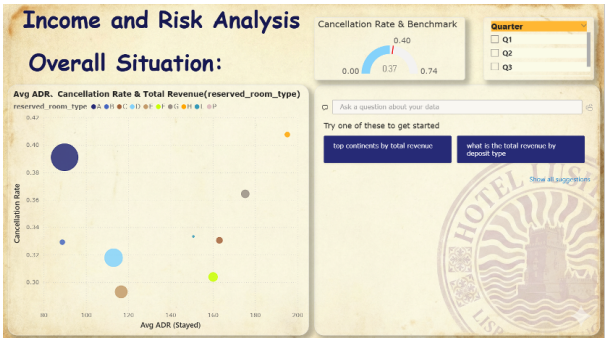


Figure A.32: Analysis of income and risks for reserved room type and Q&A Box

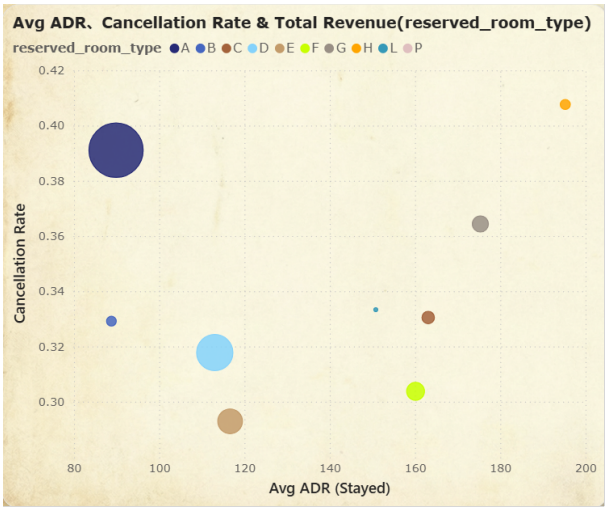


Figure A.33: Classify according to reserved room type

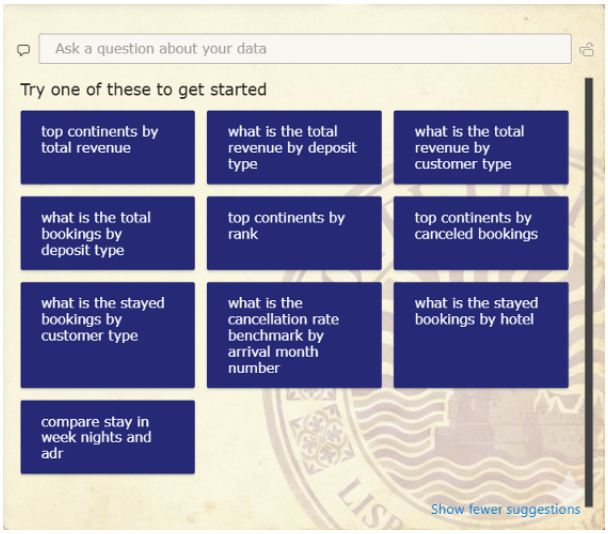


Figure A.34: Q&A Box

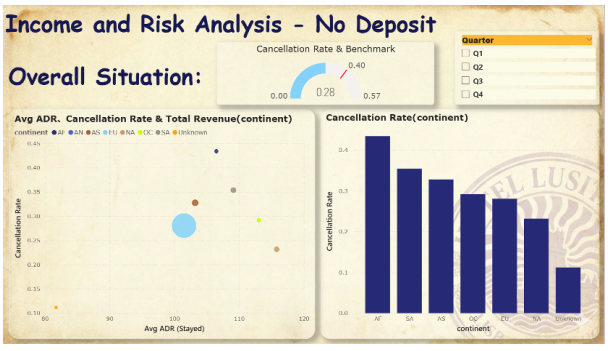


Figure A.35: Analysis of income and risks for continent - Immigration Problem

B Appendix B: Data Dictionary

Variable Name	Type	Short Description	Relevance to the Business Problem
hotel	Text (Categorical)	The type of hotel booked: City Hotel or Resort Hotel.	Distinguishes analysis between the two hotel types' booking patterns and cancellation risks.
is_canceled	Integer (Numerical)	Whether the booking was canceled (1=canceled, 0=not canceled).	Core KPI: Used to calculate the cancellation rate.
lead_time	Integer (Numerical)	Number of days between the booking date and the arrival date (advance booking time).	Risk Analysis: Used to analyze the influence of advance booking time on the cancellation rate.
arrival_date_year	Integer (Numerical)	The year of the booking arrival.	Time Intelligence: Used for year-by-year trend analysis.
arrival_date_month	Text (Categorical)	The name of the arrival month.	Time Intelligence: Used for seasonality analysis.
arrival_date_day_of_month	Integer (Numerical)	The day of the month for arrival.	Time Intelligence: Combined with year and month to

stays_in_weekend_nights	Integer (Numerical)	Number of weekend nights (Friday/Saturday) the guest stayed.	Revenue Analysis: Used to calculate total stay nights and revenue.
stays_in_week_nights	Integer (Numerical)	Number of weekday nights (Sunday to Thursday) the guest stayed.	Revenue Analysis: Used to calculate total stay nights and revenue.
adults	Integer (Numerical)	Number of adults checking in.	Demand Analysis: Used for customer segmentation.
children	Numerical	Number of children checking in.	Demand Analysis: Used for customer segmentation.
babies	Integer (Numerical)	Number of babies checking in.	Demand Analysis: Used for customer segmentation.
meal	Text (Categorical)	Type of meal booked (BB: Bed & Breakfast, FB: Full Board, HB: Half Board, SC: No Meal).	Revenue Analysis: Used to analyze demand for supplementary services.
country	Text (Categorical)	ISO code of the guest's nationality.	Geographical Analysis: Foreign Key, used to connect to the Dim_Country table.
market_segment	Text (Categorical)	Market segment channel (e.g., Online TA, Offline TA/TO, Direct, Corporate).	Risk Analysis: Used to analyze cancellation behavior across different market segments.
distribution_channel	Text (Categorical)	Distribution channel (e.g., TA/TO, Direct, Corporate).	Risk Analysis: Used to evaluate the performance and risk of different channels.
is_repeated_guest	Integer (Numerical)	Whether the guest is a repeat customer (1=Yes, 0=No).	Loyalty Analysis: Used to calculate the proportion of repeat guests.
previous_cancellations	Integer (Numerical)	Number of previous bookings canceled by the guest.	Risk Analysis: Used to assess the impact of historical cancellations on the current booking.
previous_bookings_not_canceled	Integer (Numerical)	Number of previous bookings not canceled by the guest.	Loyalty Analysis: Used to evaluate customer reliability.
reserved_room_type	Text (Categorical	The room type reserved by the	Operations Analysis: Used to analyze room

	)	customer.	type preference.
assigned_room_type	Text (Categorical)	The room type finally assigned to the customer.	Operations Analysis: Used to analyze room upgrades/downgrades .
booking_changes	Integer (Numerical)	Number of changes made to the booking before arrival.	Risk Analysis: Used to evaluate booking stability.
deposit_type	Text (Categorical)	Type of deposit (No Deposit, Non Refund, Refundable).	Risk Analysis: Used to assess the impact of deposit policies on the cancellation rate.
agent	Numerical	ID of the booking agent.	Agent Analysis: Used to analyze the contribution and efficiency of agents.
company	Numerical	ID of the booking company/entity.	Corporate Client Analysis.
days_in_waiting_list	Integer (Numerical)	Number of days the booking was in the waiting list.	Operations Analysis: Used to assess booking demand pressure.
customer_type	Text (Categorical)	Type of customer (Transient, Group, Contract, Transient-Party).	Customer Segmentation: Used to analyze spending behavior of different groups.

adr	Numerical	Average Daily Rate.	Core KPI: Used to calculate revenue and pricing trends.
required_car_parking_spaces	Integer (Numerical)	Number of required car parking spaces.	Operations Analysis: Used to assess resource requirements.
total_of_special_requests	Integer (Numerical)	Total number of special requests made.	Customer Satisfaction Analysis.
reservation_status	Text (Categorical)	The final status of the reservation (Canceled, Check-Out, No-Show).	Outcome Analysis: Provides the final transaction status.
reservation_status_date	Text (Date)	Date when the reservation status changed.	Time Intelligence: Used to analyze the timing of cancellations and check-ins.